9B Lesson 7

Bias, validity and organic farming

Year 9 Science · 9B Plant growth · Student Book pages 34 to 36

Name _____ Class _____ Date _____

Today I can:

☐ I can identify bias in graphs and arguments, explain where accidental bias comes from, and say whether a conclusion is valid.

☐ I can compare organic and intensive farming and explain biological control.

Key words

bias a shift away from the true meaning or value, on purpose or by mistake

balanced considering the points for and against something equally

systematic error an error that shifts every reading by the same amount, e.g. a wrongly printed scale

random error an error that changes from one reading to the next, e.g. stopping a clock late once

sample a small part of a whole that is measured to estimate the whole

valid does what it is supposed to do; a valid conclusion uses only the results

conclusion what the results of an investigation show

organic farming farming with natural fertilisers, very few pesticides and crop rotation

intensive farming farming with artificial fertilisers, pesticides and large fields to get the biggest yield

biological control using living organisms to reduce the number of pests

crop rotation growing different crops in a field in different years

● Core Everyone does this section.

1. Students added different numbers of drops of liquid fertiliser to bean seedlings and measured the mean increase in height after a week.

Number of drops of liquid fertiliser	2	4	6	8	10
Mean increase in height of seedlings (cm)	4.6	6.3	7.8	7.5	7.2

a) Tick one box to show a valid conclusion for this experiment.

☐ A Seedling height increases as the amount of fertiliser increases.

☐ B Too little or too much fertiliser can harm seedlings.

☐ C The more fertiliser that is added, the smaller the increase in growth.

☐ D Growth increases with more fertiliser up to a point, beyond which the amount of growth falls.

b) Give a reason for your choice.

c) State two variables that should be controlled to make this experiment valid.

2. Match each scientific term to the correct example. Draw a line.

systematic error	the stop clock was stopped late in one test during an experiment
random error	only data that support a particular view are quoted
valid experiment	a mass balance is not set to zero before the experiment
intentional bias	all other variables are controlled, so the dependent variable is only affected by the

	independent variable

3. Tick one box in each row to show whether the statement describes organic farming or intensive farming.

Statement	Organic	Intensive
mostly natural fertilisers such as manure		
many artificial pesticides		
large fields to make it easy to use big machines		
small fields with hedges and trees for predators of pests to live in		
crops are rotated so that clover adds nitrates to the soil		
the biggest yield from each field		

●● Stretch When Core is finished.

S1. A report in 2017 concluded that the mass of insects in Germany had decreased by 78% in 24 years. The data had been collected using traps that catch insects flying about a metre above the ground. Give two reasons why this conclusion may not be valid.

S2. Look at graph C on the slide (or page 34 of your book), which shows the monarch butterfly counts for 1997, 1999, 2001 and 2003 only.

a) Explain how this graph is biased.

b) Suggest who might create a graph like this, and explain your reasoning.

S3. Suggest which type of farm, organic or intensive, attracts more insect-eating birds. Explain your answer.

●●● Challenge Explain in full sentences.

C1. HT soybean is not killed by the herbicide glyphosate. Advantages: easier management of weeds; fewer sprayings needed than with other herbicides; glyphosate is not persistent. Disadvantages: expensive seed; with time, weeds unaffected by glyphosate become more common, so other herbicides may be needed; HT soybean may grow as a weed in other crops. Write a balanced paragraph to help a farmer decide whether to plant HT soybean.

C2. Two farms have the same area. One is organic and one is intensive.

a) Which farm produces the greater amount of crop? Explain your answer.

b) Explain why some people still choose to buy organic food, and state what the evidence says about whether organic food is healthier.

My thinking (any language)

A website says: 'Scientists prove that organic food is healthier.' What would you want to know about the study before you believed this claim?

Word bank bias · balanced · systematic error · random error · sample · valid · conclusion · organic farming · intensive farming · biological control · crop rotation

✂ ---------- cut here

Exit ticket 9B Lesson 7: Bias, validity and organic farming Name _____

1. What is bias?

2. Give one example of a systematic error.

3. Give one difference between organic farming and intensive farming.

Today I can: Goal 1 ☐ Goal 2 ☐